

KEIN CHAN

FULL-STACK ENGINEER

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PROFESSIONAL SUMMARY

Full-Stack Engineer and Data Architect: Over 10 years of enterprise-level full-stack development and data application experience, proficient in the full Software Development Life Cycle (SDLC), skilled in leading distributed teams, and designing efficient, stable web/mobile applications and data integration solutions to drive business objectives.

TECH STACKS

Programming :

Python, Java, Swift, Scala, C#, R

Database:

Mysql, PostgreSQL, OracleDB

Source Code Version Control:

Github, Git

Test Tools:

Pytest, Junit, Selenium, Postman

Framework:

Spring Boot, Flask, Django,

WPF,SwiftUI,React Native

Cloud Computing:

AWS,GCP,Azure

Virtualized Platforms:

Docker, Vagrant

CI/CD tools:

Travis CI, Github Actions

Data Visualization:

PowerBI, Microstrategy

UI/UX Design:

Photoshop, Figma

Project Management:

Jira

WORK EXPERIENCE

Amigo Technology Consulting Studio, Tokyo

Nov 2022 - Present

Full Stack Engineer & Data Architect

- **LY Group Data Warehouse Migration:** As the technical lead and project manager, I led the migration of the enterprise-level data platform for Japan's Line Yahoo Group, including completing the full technical solution validation for the migration from the Teradata on-premise data center to AWS Redshift; responsible for migration resource evaluation and project schedule control to ensure timely delivery; developed standardized migration processes and data validation mechanisms to ensure data consistency; coordinated technical alignment and resource allocation within internal business units of the group. Based on peer benchmarks, the project is estimated to reduce operational costs by 35%-45% (covering licenses, hardware, maintenance).
- **Japanese National Mobility Tracking System:** An internal project for SoftBank, one of Japan's three major telecom companies. By obtaining mobile base station and GPS location data, the project continuously manages the mobile phone location data, forming a data warehouse that covers approximately one-third of Japan's population. Responsible for the verification and investigation of the underlying framework and some visualization tasks. The project involves a large amount of personal information, so strict compliance with personal privacy protection requirements is essential.
- **Japan Travel Itinerary Generator App(Personal Software Development Project):** The app automatically plans travel itineraries based on user-input start and end points, modes of transportation, and travel destination information. The project includes: requirement analysis and system architecture planning, UI design, app development, backend development, admin panel development, and web scraping for data collection.
- **Nanamusic Project:** Responsible for backend management development and system maintenance, promptly addressing system failures caused by version updates, malicious attacks, etc.

Outsourcing Technology(TSE Code:2427), Tokyo**Sep 2019 – Oct 2022***Backend Engineer*

Participated as a backend engineer in the Japanese government's family registry digitization project and Toyota Group's smart city TRI-AD (now Woven Planet) development project.

- Participated in the **Japanese government's family registry digitization project** led by Fujifilm, developing five Windows applications to assist the family registry department in addressing issues in the digitalization of the family registry system. By introducing OCR (Optical Character Recognition) technology, the project converted scanned image-format family registry documents into structured data, improving processing efficiency from the previous 10 years of manual operations to just a few hours. The production environment involved Japanese citizens' family registry data and strictly adhered to Japan's Personal Information Protection Law to ensure the security and compliance of sensitive citizen data.
- Responsible for the backend development of the **smart logistics delivery system** for Toyota Group's smart city (Woven City), designing and implementing the core system based on microservices architecture to support the collaborative operation of over 3,000 delivery robots.

Oriental Kingdom Group, Tokyo**Jun 2018 – Aug 2019***Test Development Engineer*

- Responsible for testing the **Pepper humanoid robot for corporate applications** under SoftBank, leading the test plan design and execution for the Pepper management system. Worked closely with the technical team from the French subsidiary Aldebaran Robotics to coordinate multinational testing processes and track issues, ensuring the stability and functionality of the robot applications meet enterprise-level standards.

Lama Bang Technology Co., Ltd., Shenzhen**Oct 2015 – May 2018***Big Data Engineer*

Lama Bang app was one of China's largest maternity and baby social media platforms. In addition to the Lama Bang app, the company also operates apps like Pregnancy Companion, Lama Bang Mall, and Lama Bang U-Select. As a fast-growing unicorn company, Lama Bang faced the challenge of rapidly increasing data volume and how to leverage this data effectively.

- **Database Architecture Upgrade:** Participated in the design and implementation of a new-generation data warehouse system to support the data analysis needs of the company's products. Successfully addressed the data volume surge challenge brought by the increase in daily active users from 300,000 to 3.4 million, significantly improving data processing and storage efficiency.
- **Business Intelligence System Development:** Developed an enterprise-grade OSS data platform that automated the calculation and visualization of daily statistical indicators. Over 2,000 data reports were generated annually, providing real-time data support to departments like operations, product, and advertising.
- **Data-Driven Decision-Making System:** Built a user profiling system and A/B testing framework to support product iteration and user growth strategies, fostering the development of a data-driven culture within the company.

Lino-A Co, Kyoto**Apr 2014 – Sep 2015***System Engineer*

Developed a multilingual, multi-device compatible tourism information system platform using WordPress for visitors. Provided services to the Shimoda City Tourism Association and the Hikone City Tourism Association.

EDUCATION

Master Degree(MEc), Osaka University

Osaka, **Apr 2012 – Apr 2014**

Kokusyo Japanese-language school

Tokyo, **Oct 2010 – Sep 2011**

Bachelor Degree(BJ), South China University of Technology

GuangZhou, **Sep 2004 – Jul 2008**

CERTIFICATIONS

Java SE8 Gold

Linux Professional Institute Certification LPIC-1

AWS Solution Architect Associate

MySQL 8.0 Database Developer Oracle Certified Professional

MySQL 8.0 Database Administrator Oracle Certified Professional

Oracle Cloud Infrastructure 2025 Certified Generative AI Professional

Oracle Cloud Infrastructure 2025 Data Science Professional

Python Engineer Certification

Abbyy FC 12 Solution Architect Certification Advanced

IT Passport

Google Analytics Advanced

Neo4j Certified Professional

JLPT-N1

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PROJECTS

One of the most common challenges in IT project management and cross-departmental/company coordination is management inefficiencies caused by inaccurate technical and requirement estimations.

To address this, I have deliberately taken on diverse roles to broaden my expertise and minimize technical blind spots. Additionally, as a team member, I leverage my development experience to fill technical gaps and ensure project success.

LY Group Data Warehouse Migration

Technical Lead & Project Manager

Line Yahoo (LY Group), Japan's largest internet company with flagship products including Line (covering 71% population) and PayPay (66% market share), initiated this strategic migration from on-premise Teradata to AWS Redshift to handle data growth and reduce operational costs.

- **Cross-division coordination:** Orchestrated resources across business units, completing architecture assessment and migration scheduling
- **Technical research:** Conducted comprehensive analysis of Teradata-Redshift differences, establishing phased migration strategy
- **Resource evaluation:** Built a user profiling system and A/B testing framework to support product iteration and user growth strategies, fostering the development of a data-driven culture within the company
- **Standardization:** Developed SOP through pilot migration and iterative refinement
- **Data consistency:** Designed validation standards and automated verification framework with continuous optimization
- **Progress monitoring:** Real-time tracking and issue resolution throughout migration execution

Japanese National Mobility Tracking System

Big Data Engineer (Tech Lead)

This telecom analytics project processed mobile base station and GPS data to track and predict mobility patterns across Japan's 120M population, including location distribution and transportation modes.

- Led cloud platform selection (Databricks), ensuring smooth migration of legacy systems
- Developed anomaly detection and visualization tools for spatial data analysis
- Implemented data isolation and anonymization to meet strict compliance requirements

Japan Travel Itinerary Generator App

Full-stack Engineer

Independently developed an intelligent travel planning app generating personalized itineraries with navigation, transit routes and schedules based on user inputs.

- Full-cycle development from market research to launch
- Designed intuitive UI/UX using Figma
- Built iOS client with SwiftUI, integrating CloudKit and MapKit
- Developed proprietary multi-destination routing algorithm
- Created cross-platform admin system with React Native
- Implemented CI/CD pipeline with cloud infrastructure
- Implemented Agile project management via GitHub Projects, efficiently tracking task progress

Nanamusic

Server-side Engineer

As core developer for this popular Japanese music creation app, handled architecture design and DevOps using Agile methodology.

- **Server-side API Development & Maintenance:** Designed and developed RESTful API services, improved unit-testing (UT) and integration-testing (UI) pipelines, and ensured system stability during version iterations.
- **CI/CD Pipeline Optimization:** Streamlined the CI/CD workflow, cutting execution time for more than 3,600 server-side UI tests from four hours to one hour, significantly boosting development efficiency.
- **Cloud-Service Management:** Maintained AWS and other cloud platforms using Terraform-based Infrastructure as Code (IaC), reducing human error and increasing system reliability.
- **Server Monitoring & Incident Response:** Monitored server resources, promptly handled failures, rolled back releases when necessary, and mitigated malicious attacks (e.g., IP blocking).
- **Operations-Management Platform Development:** Enhanced an internal operations platform, improving the team's efficiency in handling copyright-infringement cases and ensuring timely updates to music metadata.
- **Cross-Platform Compatibility:** Resolved React Native compatibility issues during Android/iOS version upgrades to guarantee stable cross-platform performance.

Japanese Government's Family Registry Digitization (Fujifilm)

Server-side Engineer

Developed automation solutions for Japan's national registry digitization, implementing OCR and error detection to handle massive data correction workloads.

- Liaised with business stakeholders to elicit and refine requirements, participated in business-process design, and ensured technical feasibility of projects.
- Designed and implemented database schemas for multiple applications; managed more than 120 database tables.
- Devised and tuned structured-document definitions for scanned documents, abstracting document structures into relational table designs.
- Developed patch programs that raised average recognition accuracy from 65 % to over 85 %.
- Designed manual-review interfaces and streamlined human-in-the-loop workflows.
- Coordinated SDK-level technical support for the Russian ABBYY OCR engine, resolving core issues such as Japanese character recognition.
- Enforced data-compliance policies by isolating development and production datasets, safeguarding real user privacy and eliminating data-leakage risks.

Toyota Smart Logistics (Woven City)

Server-side Engineer

Developed microservices for the autonomous robot logistics network in Toyota's future smart city (2,000 residents), following Agile methodology.

- Modeled functional modules from business workflows, designed microservice entities, and delivered a highly scalable and fault-tolerant microservices architecture.
- Devised the core data-flow blueprint, defined database schemas and interface specifications, and authored comprehensive technical documentation.
- Leveraged Docker Compose for container orchestration to ensure identical and isolated dev/test/prod environments, eliminating environment-shift issues during development.
- Built REST APIs and standard microservice instances using Spring Boot, integrating MQTT and RabbitMQ for messaging.
- Designed and executed unit tests, integration tests, and functional tests to guarantee project quality.
- For external systems and internal modules still under development, rapidly created mock services with Flask to satisfy interface-testing needs during development.

Softbank Pepper Robot

Test Engineer

Conducted system testing for Pepper humanoid robot's management system and enterprise applications.

- Designed and executed end-to-end test plans for the Pepper management system and Pepper App, covering unit, integration, and functional testing at every phase.
- Led User Acceptance Testing (UAT) to guarantee the reliability and user experience of each new App release.
- Built a Selenium-based automation framework that cut manual testing effort by 87.5 % (from 4 person-days to 0.5 person-days).
- Established an automated fault-log collection system, enabling centralized logging management and analytics on AWS.
- Reproduced and analyzed newly reported bugs, gathered comprehensive logs, and generated technical reports; served as the primary technical liaison to Aldebaran Robotics in France, collaborating on hardware-compatibility issues.
- Efficiently tracked and managed the entire lifecycle of 200+ defects via the Redmine platform.

Lamabang Business Intelligence

Big Data Engineer

Built data infrastructure for China's leading maternal social platform (3.4M DAU) to enable data-driven decisions.

- Designed, defined, tested, and governed more than 3,400 tracking events; led cross-functional data-alignment sessions with business and engineering teams to translate business needs into technical solutions, enabling comprehensive user-behavior capture and analytics.
- Built high-performance ETL pipelines leveraging SQL, Shell, Python, and Scala to process massive-scale datasets.
- Architected a scalable Hadoop-based data warehouse that supported rapid growth in daily active users from 0.3 million to 3.4 million.
- Developed and maintained a real-time data-visualization platform that provides actionable insights to operations and executive stakeholders.
- Created an anomaly-detection module that alerts the ops team instantly via WeChat and email when data irregularities occur, cutting average response time by 80 %.
- Designed and executed A/B tests to deliver data-driven guidance for app development priorities.
- Established user personas by integrating machine-learning models to analyze behavioral patterns, boosting 7-day user retention by 7.67 % on average.